

Experiment: K-Means Clustering for Pulmonary Diseases Dataset

AIM

To apply the K-Means Clustering algorithm on a Pulmonary Diseases dataset and group patients into clusters based on their medical attributes.

ALGORITHM

K-Means Clustering Algorithm

Step 1: Load the pulmonary diseases dataset.

Step 2: Select relevant features such as:

- Age
- FEV1
- FVC
- Oxygen Saturation

Step 3: Normalize the data if required.

Step 4: Choose the number of clusters (K).

Step 5: Initialize K centroids randomly.

Step 6: Assign each data point to the nearest centroid.

Step 7: Recalculate cluster centroids.

Step 8: Repeat Steps 6 and 7 until cluster assignments no longer change.

Step 9: Display the clustered data and visualize the clusters.

SOURCE CODE (PYTHON PROGRAM)

```
# Import libraries
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Sample Pulmonary Diseases Dataset
data = {
    'Age': [25, 30, 45, 50, 65, 70, 28, 35, 55, 60],
    'FEV1': [3.5, 3.2, 2.8, 2.5, 1.8, 1.5, 3.4, 3.0, 2.0, 1.7],
```

```

    'FVC': [4.5, 4.2, 3.8, 3.5, 2.8, 2.5, 4.4, 4.0, 3.0, 2.7]
}

df = pd.DataFrame(data)

# Apply K-Means Clustering
kmeans = KMeans(n_clusters=3, random_state=42)
df['Cluster'] = kmeans.fit_predict(df[['FEV1', 'FVC']])

# Display clustered data
print("Clustered Dataset:")
print(df)

# Plot clusters
plt.scatter(df['FEV1'], df['FVC'], c=df['Cluster'], cmap='viridis')
plt.scatter(kmeans.cluster_centers_[:,0],
            kmeans.cluster_centers_[:,1],
            color='red',
            marker='X',
            s=200)

plt.title("K-Means Clustering for Pulmonary Diseases")
plt.xlabel("FEV1")
plt.ylabel("FVC")
plt.show()

```

INPUT

Sample Pulmonary Diseases Dataset

Age FEV1 FVC

25 3.5 4.5

30 3.2 4.2

45 2.8 3.8

50 2.5 3.5

65 1.8 2.8

Age FEV1 FVC

70 1.5 2.5

28 3.4 4.4

35 3.0 4.0

55 2.0 3.0

60 1.7 2.7

OUTPUT

Clustered Dataset:

Age FEV1 FVC Cluster

0 25 3.5 4.5 1

1 30 3.2 4.2 1

2 45 2.8 3.8 0

3 50 2.5 3.5 0

4 65 1.8 2.8 2

5 70 1.5 2.5 2

6 28 3.4 4.4 1

7 35 3.0 4.0 0

8 55 2.0 3.0 2

9 60 1.7 2

Experiment: K-Means Clustering for Pulmonary Disease Data

AIM

To apply the **K-Means Clustering Algorithm** on pulmonary disease patient data using Respiratory Rate and Oxygen Saturation values and evaluate the clustering results.

ALGORITHM

K-Means Clustering Algorithm

Step 1: Import the required libraries.

Step 2: Create the pulmonary disease dataset.

Step 3: Select the features:

- Respiratory Rate
- Oxygen Saturation

Step 4: Choose the number of clusters ($K = 2$).

Step 5: Initialize cluster centroids.

Step 6: Calculate the distance between each patient and the centroids.

Step 7: Assign each patient to the nearest cluster.

Step 8: Update the centroids by computing the mean of cluster members.

Step 9: Repeat Steps 6–8 until cluster assignments remain unchanged.

Step 10: Display the cluster assignments and visualize the clusters.

SOURCE CODE (PYTHON PROGRAM)

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans

# Pulmonary Disease Dataset
data = {
    'Patient_ID': ['P1', 'P2', 'P3', 'P4', 'P5'],
    'Age': [25, 30, 40, 45, 55],
    'Respiratory_Rate': [16, 18, 22, 24, 28],
    'Oxygen_Saturation': [98, 97, 94, 92, 88]
}

df = pd.DataFrame(data)

# Features for clustering
X = df[['Respiratory_Rate', 'Oxygen_Saturation']]

# Apply K-Means
kmeans = KMeans(n_clusters=2, random_state=42)
df['Cluster'] = kmeans.fit_predict(X)

# Display results
```

```

print("Clustered Data:")
print(df)

# Cluster centroids
print("\nCluster Centroids:")
print(kmeans.cluster_centers_)

# Visualization
plt.scatter(X['Respiratory_Rate'],
            X['Oxygen_Saturation'],
            c=df['Cluster'],
            cmap='viridis',
            s=100)

plt.scatter(kmeans.cluster_centers_[:,0],
            kmeans.cluster_centers_[:,1],
            marker='X',
            s=200,
            color='red')

for i, txt in enumerate(df['Patient_ID']):
    plt.annotate(txt,
                 (X['Respiratory_Rate'][i],
                  X['Oxygen_Saturation'][i]))

plt.title("K-Means Clustering of Pulmonary Disease Data")
plt.xlabel("Respiratory Rate (breaths/min)")
plt.ylabel("Oxygen Saturation (%)")
plt.show()

```

INPUT

Patient ID Age Respiratory Rate (breaths/min) Oxygen Saturation (%)

P1	25	16	98
P2	30	18	97
P3	40	22	94
P4	45	24	92

P5 55 28 88

Clustered Data:

Cluster Centroids:

Cluster Interpretation

Cluster 1 P1, P2 Normal respiratory condition

Cluster 0 P3, P4, P5 Possible pulmonary impairment

Thus, the **K-Means Clustering Algorithm** was successfully applied to the pulmonary disease dataset. The patients were grouped into **two meaningful clusters** based on Respiratory Rate and Oxygen Saturation values. The clustering showed good separation between healthy individuals and patients with potential respiratory disorders, demonstrating the effectiveness of K-Means for pulmonary health analysis.